

Day 25: Row-Level Security (Static & Dynamic RLS)

📌 Definition to Remember

Row-Level Security (RLS) restricts data access for given users at the dataset level, ensuring that regional managers or sales reps only see the specific rows of data they are authorized to view in a single shared report.

1. Static RLS (Hardcoded Role Rules)

- Creating fixed roles in Power BI Desktop (Modeling → Manage Roles):

```
// Role: US_East_Manager (Filter on Dim_Store)
Dim_Store[Region] = "US East"

// Role: EMEA_Sales (Filter on Dim_Geography)
Dim_Geography[Continent] = "Europe"
```

- In Power BI Service: Assign user security groups or emails to each defined role.

2. Dynamic RLS (User-Driven via `USERPRINCIPALNAME()`)

- Scales to thousands of users with a single unified security role using an **User Access Mapping Table**:

```
[UserSecurityTable]
Email                | AllowedRegion
john.doe@company.com | US East
maria.garcia@company.com | LATAM
```

- Dynamic DAX Filter on User Security Table:**

```
[ userEmail ] = USERPRINCIPALNAME()
```

- Filters propagate through relationships from the security table down to fact tables automatically.

3. Testing RLS in Desktop & Service

- Power BI Desktop*: Modeling → **View as Roles** → Enter test user email.
- Power BI Service*: Dataset Security → **Test as role**.

★ Must-Master Skills & Interview Points

- Static RLS vs Dynamic RLS: Why dynamic RLS is essential for scalable enterprise deployments.
- Functions: `USERPRINCIPALNAME()` (returns `user@domain.com`) vs `USERNAME()`.
- RLS applies strictly to users with **Viewer** permissions; Admins/Members bypass RLS.

⚡ Quick Recall

Static RLS (Fixed Roles) vs Dynamic RLS (USERPRINCIPALNAME() + Security Mapping Table)